

Session Objectives

Objective 1- **Explain** how to localize based on clinical presentation.

Objective 2- **Describe** how to exclude treatable causes of potentially fatal disorders.

Objective 3- **Correlate** the differential diagnosis of disease at each level of the motor unit.

Objective 4- **Apply** treatment options, if there are any available.

Objective 5- **Identify** a neurological emergency

Case 1

- A 15-year-old boy with anxiety, migraines and ADHD presented with rapidly progressive lower extremity weakness.
- One month prior to the weakness, he developed fever and flu-like symptoms with myalgia in both legs.
- The leg weakness persisted despite resolution of flulike symptoms.

- **Answer:** ACUTE FLACCID MYELITIS
- Flaccid limb weakness post viral infection
- Decreased muscle stretch reflexes
- Associated with pathogens: enteroviruses D68 and 71, poliovirus and flaviviruses. (Post resp or GI virus).
- More common in children. An outbreak occurs every other year.

- VITAMINS- *Differential Dx of SC lesion*
AKA- Myelopathy
- Vascular
- Infection/post-infectious
- Traumatic
- Autoimmune
- Metabolic /Toxic
- Inflammatory
- Neoplastic
- Structural



DDx for Myelopathy

- V=Spinal cord infarct, vascular malformation
- I=Virus, bacteria, fungus, **post viral**
- T=Decompression illness, trauma
- A=Autoimmune- Lupus, Sjogren's, MS, NMO, MOG
- M=B12 deficiency, drug exposures, nitrous oxide toxicity
- I= Neurosarcoidosis
- N=Spinal cord tumors- primary or metastatic
- S=Syrinx or other congenital anomalies of the cord

Symptoms of Myelopathy

- Depends on the tracts affected and the disease process.
 - Gait instability/imbalance
 - Stiffness or clumsiness
 - Weakness in the arms or legs
 - Urinary urgency /bowel incontinence-
 - Sensory level- sensitive squeezing belt, irritating bandlike sensation at the superior margin and numbness or tingling below the level.
 - Episodic muscle spasms, which cause unintentional extension of legs or cramping in the calves

Signs of a myelopathy

- Flaccid paralysis (corticospinal tract) evolves to
- Weakness with increased tone, brisk DTRs
- Babinski sign
- Hoffman sign (if cervical cord is involved). Fingers and thumb flex involuntarily to a stimulus.

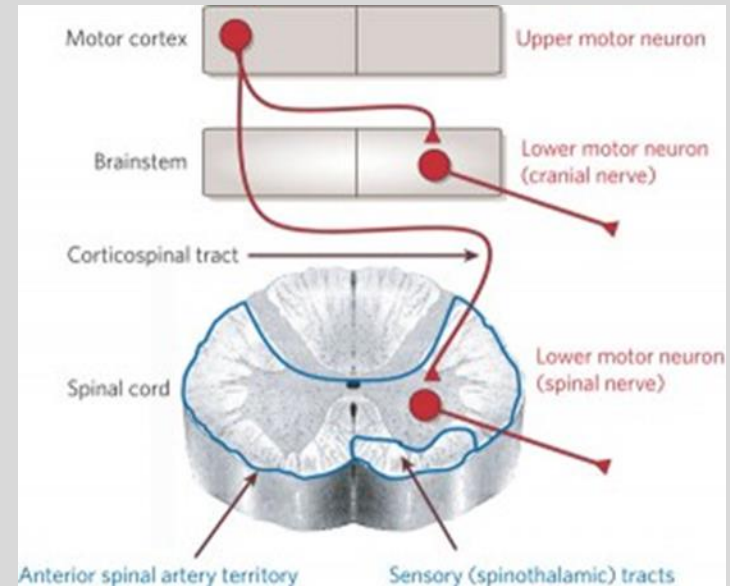
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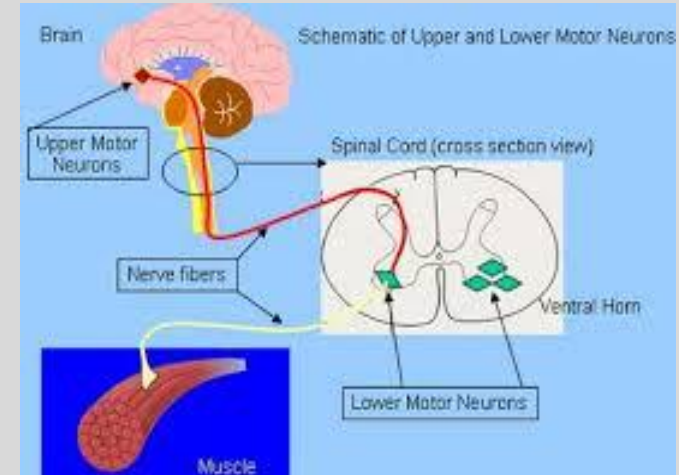
Upper motor neurons

- UMN's originate in the primary motor cortex of the brain.
- Their axons comprise the **corticobulbar tract** (connecting to the brainstem) and the **corticospinal tract** (connecting to the spinal cord).



Lower motor neurons

- Also referred to as **alpha motor neurons** or **anterior horn cells**.
- Located in motor nuclei in the brainstem or the anterior gray matter of the spinal cord.
- Axons connect to muscles of the bulbar region or limbs.



Anterior Horn Syndrome

- Refers to lower motor neuron degeneration
- Weakness
- Preservation of all sensory modalities
- Examples:
 - Spinal muscular atrophy
 - Spinal and bulbar muscular atrophy (Kennedy disease)
 - **West Nile myelitis**
 - Poliomyelitis

Spinal Muscular Atrophy

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- The spinal muscular atrophies are a group of inherited disorders that cause degeneration of anterior horn cells.
- Most common type is caused by a variation in the SMN (survival motor neuron 1).
- There are now 9 recognized types identified. All involve varying degrees of weakness
- Clinical presentation varies from early death in infancy (type 0, 1A) to walking independently as an adult (type 4)

Poliomyelitis

- A highly infectious enterovirus.
- 1% of infected patients will develop acute paralytic syndrome due to infection of the anterior horns cells.
- Painful muscle spasms occur before flaccid weakness (spinal and/or bulbar)
- Vaccination remains the main means of polio prevention.
- Iron lung (1940s-early 1950s)/vaccine 1955

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Case 3

- A 67-year-old man presented with an 8-month history of slurred speech.
- Over that same time period he had weakness of the left hand without numbness.
- Cramping was noted in his hands and legs.
- He admitted to crying spontaneously.

Clinical features of ALS

Motor symptoms

- Weakness of limbs with atrophy
- Bulbar (speaking, chewing, swallowing)
- Brisk reflexes

Pseudobulbar affect

- Laughing or crying- not under voluntary control

Cognitive features

- Small proportion may also have frontotemporal dementia

ALS

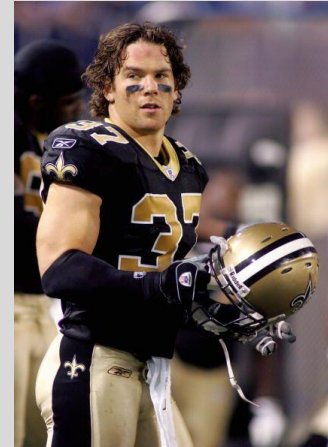
- Combination of upper and lower motor neuron symptoms.
- LMN- **weakness, fasciculations, atrophy**
- UMN-**spasticity, hyperreflexia, and pathologic reflexes** (i.e. Babinski).
 - Note: A retained reflex in a weak limb is considered ***abnormal*** and is a clinical clue.
 - Pelvic floor- urinary and rectal sphincters are unaffected.

Dx ALS

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- Combination of history, exam findings, electrophysiologic features. Exclude other causes.



Images: Wikipedia

Continuum- Lifelong Learning in Neurology. Peripheral Nerve and Motor Neuron Disorders. October 2023 VOL. 29 NO. 5

Features of a DRG lesion

- *Non-length dependent, asymmetric sensory deficits and preserved motor strength*
- Proximal areas may be affected prior to distal areas. This includes upper extremities, face and trunk.
- Pain is common.
- Profound sensory loss with gait and limb ataxia.
- Absent deep tendon reflexes

Nerve Root- DDx

- Most common- intervertebral disk herniation and spondylosis.
- Synovial cysts (less common)
- Infection (osteomyelitis and diskitis)
- Vertebral fracture
- Bony malignant disease
- Epidural hematoma

Symptoms of a Radiculopathy

- Radiating pain
- Reduced reflex along spinal root.
- Sensory loss may be present.
- Muscles weakness supplied by corresponding nerves.

Neuralgic Amyotrophy- Parsonage-Turner syndrome (a type of plexopathy)

- Typically presents with acute onset of shoulder pain (unilateral).
- Pain resolves, but then evolves into sensory loss and weakness of the affected limb.
- Risk factors: recent viral illness, surgery
- This is inflammatory in nature.

Distal symmetric polyneuropathy

- Diabetes or prediabetes-most common
- Infectious Neuropathies
- Chemotherapy
- Chronic alcohol use and other toxins
- Vitamin B12 deficiency (uncommon)
- Hereditary- hereditary motor sensory neuropathy
- Entrapment (carpal tunnel syndrome)-
mononeuropathy

Guillain-Barre Syndrome

- Encompasses a group of related disorders:

Acute inflammatory demyelinating polyradiculoneuropathy (AIDP) is the most common form.

- Post-infectious and paralytic
- CMV, Mycoplasma pneumonia, EBV, influenza A, H. flu.
- Campylobacter jejuni (causes GI) is most frequently associated bacteria.
- GBS has been found to occur after Zika virus and SARS COVID-19 infections.

Why ICU care is necessary

- Dysautonomia
- Bulbar dysfunction
- Severe or rapidly worsening weakness
- Evolving respiratory distress (inability to count 15 or more during a single breath)
- Rapid decline in respiratory function (>30% in 24 hours)

Treatment of GBS

- Supportive care is the cornerstone of treatment. Evidence based research has shown that
- **IVIG (IV immunoglobulin)** and
- **Plasma Exchange** are equally efficacious.
- Either one should be started ASAP once dx is confirmed.
- Steroids are **not** effective

CIDP

- Symmetric proximal and distal weakness and sensory loss develop over at least **2 months** in a progressive, stepwise or relapsing course.
- Reduced or absent reflexes.
- Usually no antecedent illness.
- Less common dysautonomia and respiratory failure
- Frequent postural tremor

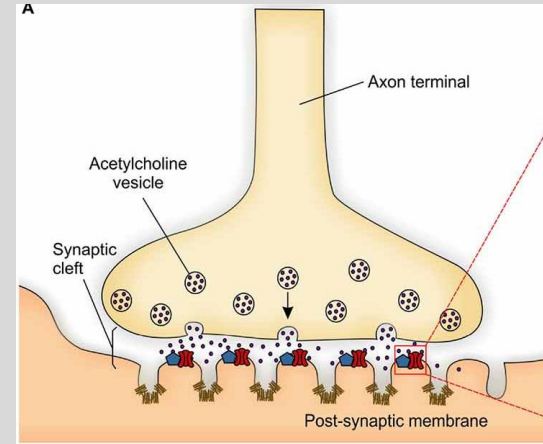
Treatment for CIDP

- Treated with one of 3 first line therapies:
 - Immunoglobulins- IV or SQ
 - Corticosteroids
 - Plasma Exchange

50-70% respond to one of these therapies

DDx of NMJ Disorders

- Myasthenia Gravis
- Lambert Eaton Myasthenic Syndrome
- Botulism
- Immune Checkpoint Inhibitor-related MG

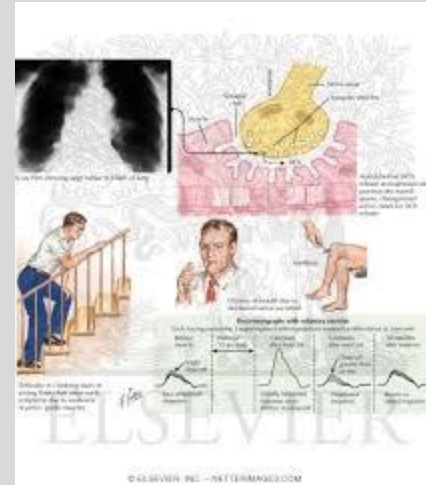


Autoimmune Myasthenia Gravis

- Autoimmune disease- causing fatiguable ocular, limb and bulbar muscle weakness-to different degrees.
- Peaks in younger women and older men.
- Antibodies against AChR or MuSK in the majority of patients.
- Highly treatable with immunomodulation.
- Thymectomy may be warranted.

Lambert Eaton Myasthenic Syndrome

- Rare autoimmune disease that affects **presynaptic** neuromuscular transmission.
- Antibodies against voltage gated calcium channels.
- Half of patients with small cell lung cancer.
- Proximal muscle weakness, hyporeflexia/areflexia and dysautonomia.
- Weakness and DTRs briefly improves after exercise.
- Treatment: 3,4-diaminopyridine/ oncologist for CA



Botulism

- A paralytic illness caused by Botulinum neurotoxins, *produced by a strain of Clostridium botulinum:inhibits ACh release at the neuromuscular junction.*
- Acute descending flaccid paralysis.
- Enters the body via: ingestion (foodborne), wound colonization, intestinal colonization (infant and adult), cosmetic or aerosolized.
- Blurred vision, slurred speech, descending paralysis, resp failure.

Honey may
quiet them,
but botulism
can kill them



Differential of Muscle Disorders

- Inherited myopathies
 - Dystrophinopathies
- Inflammatory Myopathies
 - Inclusion body
 - Polymyositis
- Acquired myopathies
 - Immune checkpoint inhibitor
 - Statin induced
 - Thyroid ophthalmopathy

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- ## On May 21st, show this child you care

For every basket of Kentucky Fried Chicken® ordered on Buckle Up! Buckle Up! Buckle Up! Buckle Up!

We want nothing more, than a little help from the little ones you're raising. And these kids don't need anything from companies of whose reputations are beyond question.

When you order a bucket of chicken on May 21st you get a giving hand on their giving nature and their America.

After each every family, Buckle Up! Buckle Up! Buckle Up! Buckle Up!

So the little ones can see what's in it for them, you can help them feel what's in it for them.

With a special offer on May 21st, the first time you order a bucket of chicken, Kentucky Fried Chicken becomes the only place where you can get a little help from the little ones you're raising.

After each every family, Buckle Up! Buckle Up! Buckle Up! Buckle Up!

Buckle Up!

hope day

Summary Slide

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- Summary Point 1: Localize the lesion based on clinical presentation and generate a differential diagnosis based on the localization.
- Summary Point 2: Do not miss a treatable diagnosis and know what the appropriate treatment is.
- Core References:
 - Continuum- Lifelong Learning in Neurology. Muscle and Neuromuscular Junction Disorders. October 2025 VOL. 31 NO. 5
 - Continuum- Lifelong Learning in Neurology. Peripheral Nerve and Motor Neuron Disorders. October 2023 VOL. 29 NO. 5.
 - Continuum- Lifelong Learning in Neurology. Spinal Cord Disorders. February 2024 Volume 30 No.1